

## WILLIAMSBURG JAMESTOWN, VA, USA

WMO: 720297

Lat: 37.239N

Lon: 76.716W

Elev: 49

StdP: 14.67

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03730

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 18.9       | 24.7 | 1.3                           | 5.9 | 23.4 | 7.1 | 7.9 | 28.6 | 23.2                  | 47.5 | 20.0 | 45.5 | 4.7                   | 340  | 0.386 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 18.5                   | 97.3            | 78.4 | 93.5 | 76.7 | 91.4 | 76.0 | 81.3                | 91.2 | 79.9 | 89.7 | 78.6 | 88.0 | 5.7                  | 150  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 79.1                            | 151.4 | 85.4 | 77.2 | 141.9 | 83.9 | 75.4 | 133.3 | 82.4 | 44.7          | 89.8 | 43.2 | 89.2 | 42.2 | 88.2 | 86.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%   | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |
|                   |      |      | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 18.1              | 15.6 | 12.6 | 13.2                       | 101.5 | 5.4                | 2.8 | 9.3                                                | 103.5 | 6.1        | 105.1 | 3.1        | 106.7 | -0.8       | 108.7 |
|                   |      |      | 11.2                       | 83.1  | 4.9                | 1.8 | 7.7                                                | 84.4  | 4.8        | 85.4  | 2.1        | 86.4  | -1.5       | 87.6  |
|                   |      |      | DB                         |       |                    |     |                                                    |       |            |       |            |       |            |       |
|                   |      |      | WB                         |       |                    |     |                                                    |       |            |       |            |       |            |       |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 62.4  | 41.3  | 44.2  | 51.1  | 62.0  | 70.7  | 78.4  | 82.1  | 80.6  | 74.9  | 64.1  | 52.4  | 46.1 | (6)  |
|      |                                                                   | DBStd       | 16.41 | 10.38 | 10.05 | 10.02 | 8.64  | 7.74  | 5.63  | 4.46  | 4.42  | 5.82  | 8.04  | 8.24  | 9.08 | (7)  |
|      |                                                                   | HDD50       | 899   | 306   | 218   | 112   | 7     | 0     | 0     | 0     | 0     | 3     | 68    | 185   |      | (8)  |
|      |                                                                   | HDD65       | 3051  | 734   | 586   | 444   | 159   | 36    | 0     | 0     | 2     | 116   | 386   | 588   |      | (9)  |
|      |                                                                   | CDD50       | 5431  | 38    | 54    | 146   | 366   | 642   | 852   | 995   | 948   | 747   | 439   | 139   | 65   | (10) |
|      |                                                                   | CDD65       | 2108  | 0     | 2     | 13    | 68    | 213   | 402   | 530   | 483   | 299   | 88    | 7     | 3    | (11) |
|      |                                                                   | CDH74       | 20210 | 1     | 23    | 130   | 655   | 1827  | 4001  | 5800  | 4782  | 2359  | 589   | 32    | 11   | (12) |
|      |                                                                   | CDH80       | 8668  | 0     | 1     | 22    | 208   | 696   | 1796  | 2794  | 2133  | 875   | 141   | 2     | 0    | (13) |
| (14) | Wind                                                              | WSAvg       | 4.6   | 5.6   | 6.0   | 6.2   | 6.0   | 4.5   | 4.1   | 3.5   | 3.2   | 3.3   | 4.2   | 4.4   | 4.7  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 48.2  | 3.9   | 3.4   | 4.3   | 3.2   | 4.3   | 3.8   | 5.6   | 5.1   | 4.6   | 3.3   | 3.2   | 3.4  | (15) |
| (16) |                                                                   | PrecMax     | 75.6  | 8.1   | 8.1   | 9.3   | 6.5   | 9.4   | 10.0  | 12.0  | 13.8  | 21.3  | 10.0  | 8.0   | 6.7  | (16) |
| (17) |                                                                   | PrecMin     | 29.6  | 0.6   | 0.8   | 0.2   | 0.7   | 0.9   | 0.6   | 0.6   | 1.3   | 0.6   | 0.0   | 0.4   | 0.4  | (17) |
| (18) |                                                                   | PrecStd     | 10.0  | 1.7   | 1.6   | 1.9   | 1.4   | 2.0   | 2.0   | 2.6   | 3.1   | 4.1   | 2.2   | 1.9   | 1.5  | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% DB     | 72.6  | 78.7  | 82.3  | 90.5  | 93.5  | 99.7  | 101.7 | 99.2  | 96.5  | 89.8  | 77.5  | 75.1  |      | (19) |
| (20) |                                                                   | 0.4% MCWB   | 62.9  | 65.5  | 65.2  | 69.4  | 74.8  | 77.6  | 78.7  | 79.4  | 76.0  | 74.3  | 65.1  | 67.0  |      | (20) |
| (21) |                                                                   | 2% DB       | 67.7  | 72.0  | 78.6  | 85.7  | 90.6  | 96.5  | 97.4  | 95.2  | 91.2  | 83.9  | 73.1  | 70.3  |      | (21) |
| (22) |                                                                   | 2% MCWB     | 60.3  | 60.2  | 62.3  | 67.5  | 73.5  | 77.0  | 79.6  | 77.8  | 75.1  | 72.4  | 63.1  | 65.1  |      | (22) |
| (23) |                                                                   | 5% DB       | 63.1  | 66.0  | 72.6  | 81.5  | 88.0  | 91.5  | 94.8  | 92.6  | 89.6  | 80.9  | 70.3  | 64.4  |      | (23) |
| (24) |                                                                   | 5% MCWB     | 56.6  | 58.4  | 59.5  | 65.1  | 72.8  | 75.4  | 78.1  | 76.7  | 74.8  | 70.1  | 62.7  | 60.4  |      | (24) |
| (25) |                                                                   | 10% DB      | 57.3  | 61.2  | 68.0  | 77.1  | 83.7  | 90.0  | 91.3  | 90.3  | 84.3  | 77.0  | 66.0  | 61.2  |      | (25) |
| (26) |                                                                   | 10% MCWB    | 51.5  | 52.4  | 58.0  | 64.2  | 70.8  | 75.0  | 77.0  | 76.2  | 72.4  | 68.3  | 59.9  | 56.3  |      | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% WB     | 65.8  | 66.9  | 68.1  | 74.0  | 77.5  | 81.1  | 84.1  | 82.7  | 79.4  | 77.5  | 69.7  | 69.9  |      | (27) |
| (28) |                                                                   | 0.4% MCDB   | 69.8  | 74.5  | 78.0  | 81.5  | 89.7  | 92.6  | 95.9  | 91.9  | 89.8  | 82.4  | 73.9  | 72.9  |      | (28) |
| (29) |                                                                   | 2% WB       | 61.8  | 63.4  | 65.3  | 70.8  | 75.6  | 79.2  | 81.7  | 81.0  | 78.1  | 75.1  | 66.8  | 65.2  |      | (29) |
| (30) |                                                                   | 2% MCDB     | 66.8  | 67.4  | 72.5  | 80.5  | 86.1  | 91.3  | 92.5  | 89.5  | 87.1  | 80.2  | 71.3  | 68.2  |      | (30) |
| (31) |                                                                   | 5% WB       | 57.6  | 59.2  | 62.8  | 68.2  | 74.3  | 77.9  | 80.2  | 79.5  | 77.0  | 73.2  | 64.0  | 62.1  |      | (31) |
| (32) |                                                                   | 5% MCDB     | 62.2  | 65.4  | 70.1  | 77.8  | 83.1  | 88.8  | 90.6  | 87.9  | 84.7  | 78.1  | 67.7  | 64.6  |      | (32) |
| (33) |                                                                   | 10% WB      | 53.3  | 53.7  | 59.5  | 65.8  | 72.9  | 76.7  | 78.9  | 78.3  | 75.6  | 70.2  | 61.3  | 57.1  |      | (33) |
| (34) |                                                                   | 10% MCDB    | 56.8  | 59.4  | 66.1  | 74.2  | 80.7  | 86.2  | 88.5  | 86.6  | 81.9  | 75.2  | 65.9  | 60.4  |      | (34) |
| (35) | Mean Daily Temperature Range                                      | MDBR        | 18.1  | 19.9  | 20.8  | 21.8  | 19.8  | 19.2  | 18.5  | 17.9  | 18.1  | 18.8  | 19.6  | 18.1  |      | (35) |
| (36) |                                                                   | 5% DB MCDBR | 24.2  | 25.7  | 25.9  | 25.4  | 22.8  | 21.2  | 20.8  | 19.9  | 20.9  | 20.8  | 22.8  | 21.2  |      | (36) |
| (37) |                                                                   | 5% DB MCWBR | 18.4  | 17.2  | 14.0  | 10.6  | 9.5   | 7.7   | 7.1   | 6.6   | 7.3   | 10.1  | 14.1  | 16.8  |      | (37) |
| (38) |                                                                   | 5% WB MCDBR | 22.5  | 24.9  | 22.5  | 21.2  | 19.4  | 19.0  | 18.8  | 17.5  | 16.8  | 16.4  | 19.1  | 19.8  |      | (38) |
| (39) |                                                                   | 5% WB MCWBR | 19.4  | 18.9  | 15.0  | 10.9  | 9.0   | 7.8   | 7.4   | 7.0   | 7.0   | 9.0   | 14.7  | 17.2  |      | (39) |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.304 | 0.316 | 0.345 | 0.391 | 0.432 | 0.486 | 0.515 | 0.511 | 0.408 | 0.352 | 0.321 | 0.308 |      | (40) |
| (41) |                                                                   | taud        | 2.537 | 2.496 | 2.448 | 2.329 | 2.245 | 2.143 | 2.079 | 2.078 | 2.352 | 2.507 | 2.546 | 2.559 |      | (41) |
| (42) |                                                                   | Ebn at Noon | 283   | 292   | 292   | 282   | 271   | 256   | 247   | 245   | 267   | 275   | 274   | 274   |      | (42) |
| (43) |                                                                   | Edh at Noon | 24    | 29    | 33    | 40    | 44    | 49    | 51    | 50    | 36    | 28    | 24    | 23    |      | (43) |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 740   | 979   | 1301  | 1648  | 1863  | 2013  | 1917  | 1730  | 1404  | 1110  | 835   | 656   |      | (44) |
| (45) |                                                                   | RadStd      | 60    | 108   | 105   | 122   | 168   | 136   | 94    | 89    | 126   | 130   | 69    | 53    |      | (45) |

## Historical Trends

|                         |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (73 neighbors) |  | +0.85 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +299  | +191  |

Nomenclature: See separate page